

Date: 16. Dec. 2019

Dear Sirs and Madams,

Cable routing, grounding, and shielding on motor yachts

The BNVG automation system was designed and approved with an electrically isolated power supply to fulfill the requirements of the classification societies. Therefore, MTU recommends to setup the supply voltage electrically isolated from the grounding/shielding (see page 7 of the Functional Description E531439_10E).

To operate the BlueVision_Basic | NewGeneration system on a vessel in which the negative pole of the battery is connected to vessel ground (e.g. with single-pole battery-charging generator), the following conditions must be met (see Fig. 1 on page 2):

- Unit grounding must be achieved by installation of a separate grounding strap
- The MTU system components must be connected with a forward and return conductor (+24V & GND). It is not allowed to locally connect a component's minus to the chassis (ships hull) as is common practice in the car industry.
- It is recommended to use a busbar / a low-impedance combination of busbars as a GND reference for the MTU system components.
- Third-party systems that are electrically connected to the BVNG system must be connected via a joint GND busbar.

Failure to observe these instructions may have the following consequences:

- The shielding effect for all cables and devices is no longer provided.
- External interferences will not be discharged and may affect the system.
- If faults occur the troubleshooting will be much more complex and might require more time.
- In the event of an unwanted ground fault, cross currents might occur, which could considerably affect or even damage devices.

When connecting Battery-minus with ship's hull, the customer must be aware that:

- in the event of contact of a single core of our system with the ship's hull a failure would immediately occur, which - when Battery-minus with ship hull is separated - would have no effect. A connected ground fault monitoring system would indicate the fault, which can be corrected before there is a negative impact on the vessels operation.
- Uncontrollable currents through the metallic hull create a more or less strong potential difference which can have an impact on the vessel's functionality.

Kind regards,



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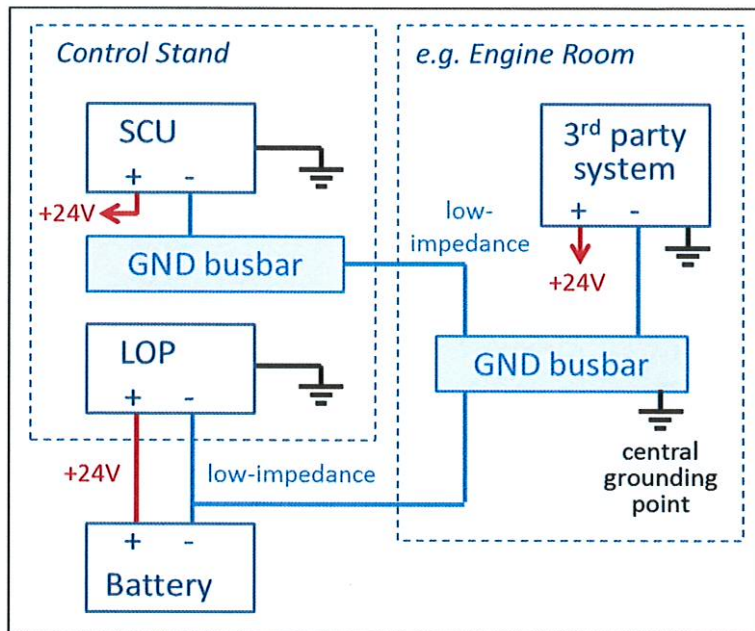


Fig.1